



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

035033-Intro to Sensor-Integrated Systems

Lecture III: Dynamic Characteristics of Measurement Systems

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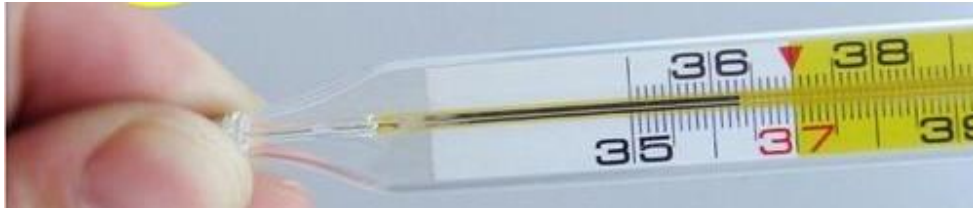
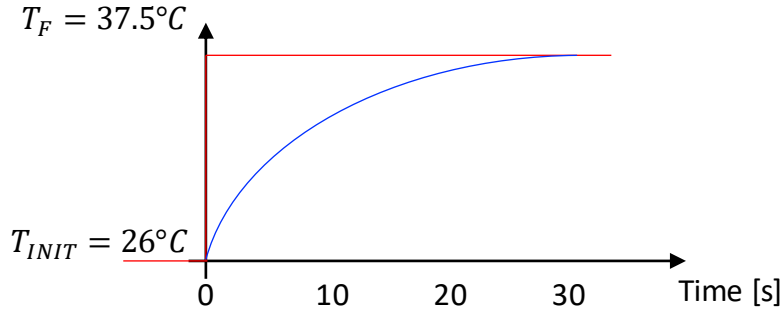
Important features of sensors

6. **Noise floor:** output variation with constant input (limits smallest usable signal). If the input is 0°C , reading is 0.01°C or 0.001°C
7. **Bandwidth / Update rate (dynamic response):** frequency range (or max update rate) the sensor can track to spec. $1/300\text{s}=0.003\text{ Hz}$
8. **Drift / Long-term stability:** output change over time with constant input. 10°C input for 1 day, measurement is 10.001°C , drift is $0.001^{\circ}\text{C/day}$
9. **Linearity:** max deviation from a straight-line input $\rightarrow$ output relation (often %FS (Full Span)). Length is proportional to temperature, but slightly nonlinear.
10. **Cross-sensitivity / Crosstalk (selectivity):** unwanted response to “other” inputs (other axes, temperature, EM fields, etc.). If 10°C with electromagnetic interference (EMI) 3 V/m , measurement is 10.001°C , then cross-sensitivity is $0.0003^{\circ}\text{C}/(\text{V/m})$



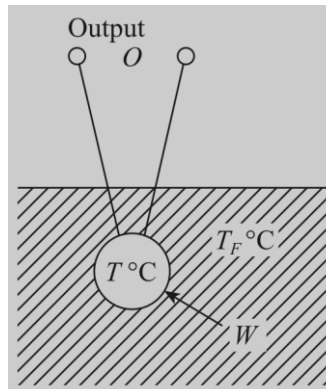
Dynamic Characteristics of Measurement Systems

- The ways in which an element responds to sudden input changes are termed its **dynamic characteristics**



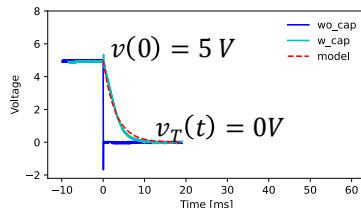
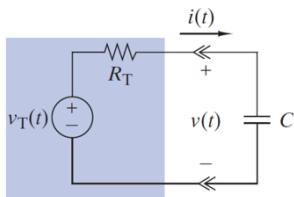
Dynamic Characteristics of Measurement Systems

- ❑ Energy balance: energy accumulation = in flow – out flow
- ❑ Initial temperature T_{INIT} [$^{\circ}C$] and environment temperature T_F [$^{\circ}C$]
- ❑ Assume the real time temperature T [$^{\circ}C$]
- ❑ In flow: $W = UA(T_F - T)$ [Watts], where U [$\frac{W}{m^2^{\circ}C}$] is the heat transfer coefficient between the fluid and the sensor, A [m^2] is the effective transfer area
- ❑ Energy accumulation $MC(T - T_{INIT})$ [J], where M [kg] is the sensor mass, C [$\frac{J}{kg^{\circ}C}$] is the specific heat of the sensor material
- ❑ M , C , and T_{INIT} are constants, then $\frac{dMC(T - T_{INIT})}{dt} = MC \frac{dT}{dt}$ [W]
- ❑ $MC \frac{dT}{dt} = UA(T_F - T) \Rightarrow \frac{MC}{UA} \dot{T} + T = T_F$
- ❑ Homogeneous solution $T(t) = K_1 e^{-\frac{UA}{MC}t}$
- ❑ Particular solution: $T(t) = T_F$
- ❑ Complete solution: $T(t) = K_1 e^{-\frac{UA}{MC}t} + T_F$
- ❑ Initial state $T(0) = T_{INIT} \Rightarrow T(t) = (T_{INIT} - T_F)e^{-\frac{UA}{MC}t} + T_F$
- ❑ Time constant $\tau = \frac{MC}{UA} \Rightarrow T(t) = (T_{INIT} - T_F)e^{-\frac{t}{\tau}} + T_F$



First order system in Mechatronics

- ❑ KVL: $v_T(t) - R_T C \dot{v}(t) - v(t) = 0 \Rightarrow R_T C \dot{v}(t) + v(t) = v_T(t)$
- ❑ Homogeneous solution $R_T C \dot{v}(t) + v(t) = 0 \Rightarrow v(t) = A e^{-\frac{1}{R_T C} t}$
- ❑ Particular solution: $R_T C \dot{v}(t) + v(t) = 0 \Rightarrow v(t) = 0$
- ❑ Complete solution: $v(t) = A e^{-\frac{1}{R_T C} t}$
- ❑ Initial condition $v(t = 0) = 0 \text{ V} \Rightarrow v(t) = 5 e^{-\frac{t}{R_T C}}$
- ❑ Time constant $\tau = R_T C \Rightarrow v(t) = 5 e^{-\frac{t}{\tau}}$



Exercise

4.1

A temperature measurement system consists of linear elements and has an overall steady-state sensitivity of unity. The dynamics of the system are determined by the first-order transfer function of the sensing element. At time $t = 0$, the sensing element is suddenly transferred from air at $20\text{ }^{\circ}\text{C}$ to boiling water. One minute later the element is suddenly transferred back to air. Using the data given below, calculate the system dynamic error at the following times: $t = 10, 20, 50, 120$ and 300 s .

Sensor data

Mass = $5 \times 10^{-2}\text{ kg}$

Surface area = 10^{-3} m^2

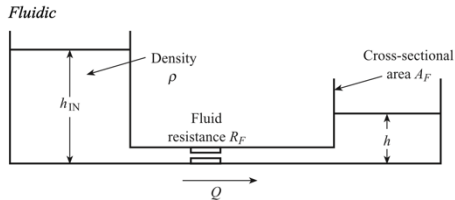
Specific heat = $0.2\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

Heat transfer coefficient for air = $0.2\text{ W m}^{-2}\text{ }^{\circ}\text{C}^{-1}$

Heat transfer coefficient for water = $1.0\text{ W m}^{-2}\text{ }^{\circ}\text{C}^{-1}$

Analogous first-order elements

- ❑ System can be in different domains
- ❑ But equation of motion is the “same”
- ❑ $\frac{A_F R_F}{\rho g} \dot{h} + h = h_{IN}$
- ❑ $RC \dot{V} + V = V_{IN}$
- ❑ $\frac{\lambda}{k} \dot{F} + F = F_{IN}$
- ❑ Better way: $\lambda \dot{x} + kx = 0$



Volume flow rate $Q = \frac{1}{R_F} (P_{IN} - P)$

Pressures $P_{IN} = h_{IN} \rho g$, $P = h \rho g$

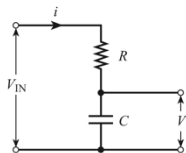
$A_F \frac{dh}{dt} = Q = \frac{\rho g}{R_F} (h_{IN} - h)$

$\frac{A_F R_F}{\rho g} \frac{dh}{dt} + h = h_{IN}$

i.e.

$\tau_F \frac{dh}{dt} + h = h_{IN}$, $\tau_F = \frac{A_F R_F}{\rho g}$

Electrical



$V_{IN} - V = iR$

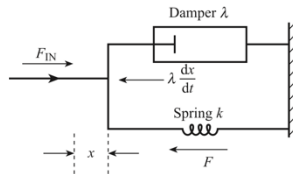
Charge $q = CV$, current $i = \frac{dq}{dt} = \frac{CdV}{dt}$

$RC \frac{dV}{dt} + V = V_{IN}$

i.e.

$\tau_E \frac{dV}{dt} + V = V_{IN}$, $\tau_E = RC$

Mechanical



$F_{IN} - F = \lambda \frac{dx}{dt}$, $\lambda \text{ N s m}^{-1} = \text{damping constant}$

Displacement $x = \frac{F}{k}$, $k \text{ N m}^{-1} = \text{spring stiffness}$

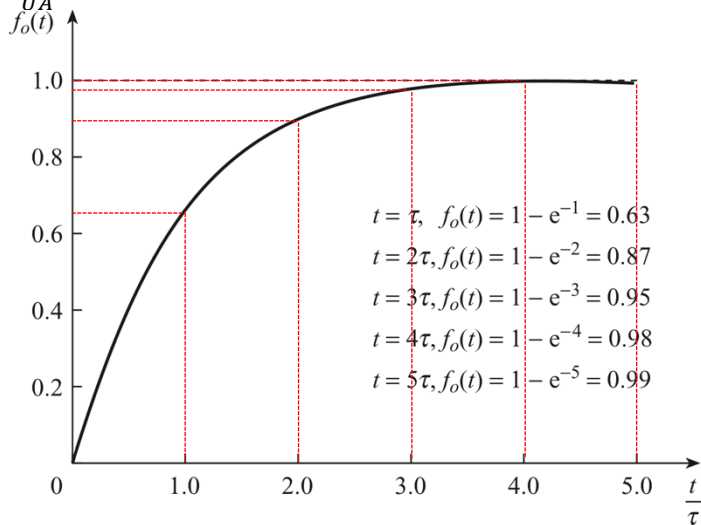
$\frac{\lambda}{k} \frac{dF}{dt} + F = F_{IN}$

$\tau_M \frac{dF}{dt} + F = F_{IN}$, $\tau_M = \frac{\lambda}{k}$

Step response of first- and second-order elements

□ $T(t) = (T_{INIT} - T_F)e^{-\frac{UA}{MC}t} + T_F \Rightarrow T(t) = (T_{INIT} - T_F)e^{-\frac{t}{\tau}} + T_F$

□ Time constant $\tau = \frac{MC}{\frac{UA}{f_o(t)}}$: the time needed to reach $1 - e^{-1} = 63\%$ of the full capacity



Intuitive understanding of time constant

- ❑ Time constant: 2 hours vs 63 hours
- ❑ Time constant larger $\rightarrow$ slower
- ❑ For sensors, the smaller, the better

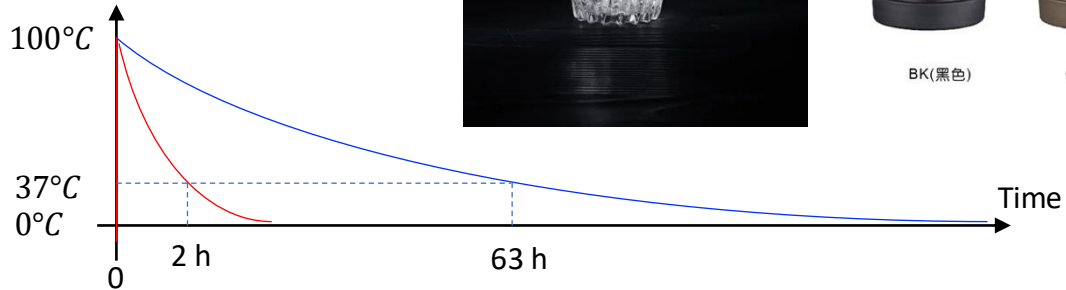


THERMOS. 膳魔師不銹鋼真空保溫杯
TCMF-501 500ml

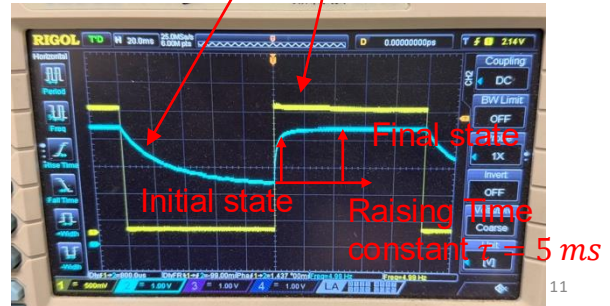
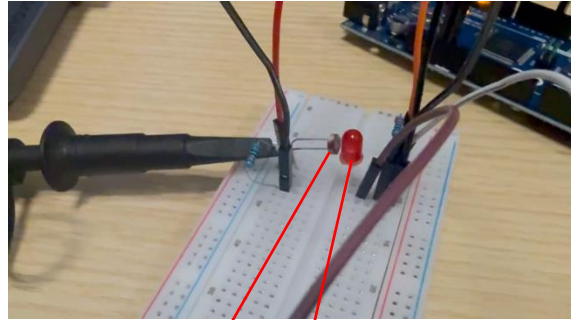
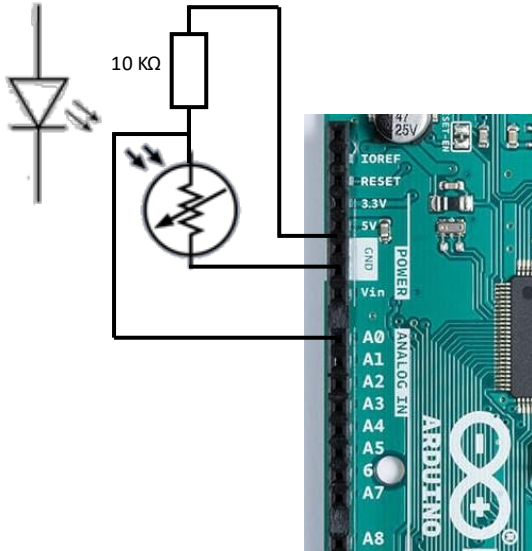


BK(黑色)

GL(金色)



Bandwidth: explain intuitively



Bandwidth: explain intuitively

Falling Time constant $\tau = 70 \text{ ms}$

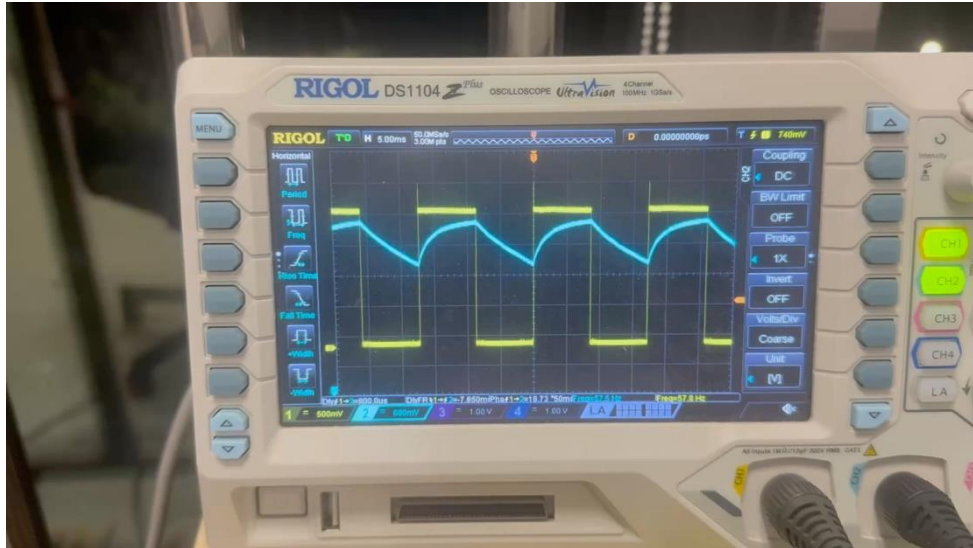
Initial state

Final state



Example

- ❑ Beyond 70 ms, *i.e.*, $1000 \text{ ms} / (2 * 70 \text{ ms}) \approx 7 \text{ Hz}$, system is not reliable.
- ❑ Higher frequency (1000 Hz), the system does not work



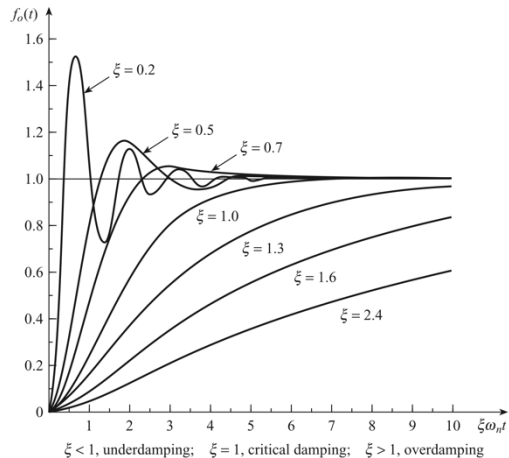
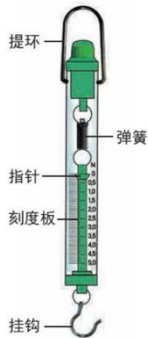
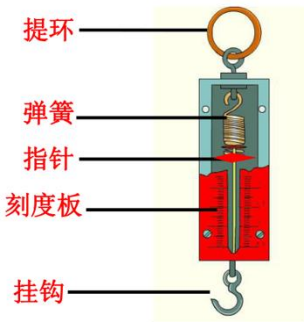
Infrared Thermal Camera

- ❑ Time constant < 5 ms



Example of second order system

弹簧测力计的基本构造



Example of second order system

❑ Question: can you do 100 measurements within 1 second?



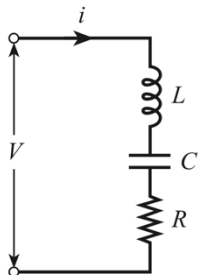
Second order system

❑ Spring-mass-damper system: $m\ddot{x} + c\dot{x} + kx = F$

❑ Electric circuit $L\ddot{q} + R\dot{q} + \frac{1}{C}q = V$

❑ Nondimensionlized form: $\ddot{x} + 2\omega_n\zeta\dot{x} + \omega_n^2x = \frac{F}{m}$, where resonance frequency $\omega_n = \sqrt{\frac{k}{m}}$,
damping ratio $\zeta = \frac{c}{2m\omega_n}$

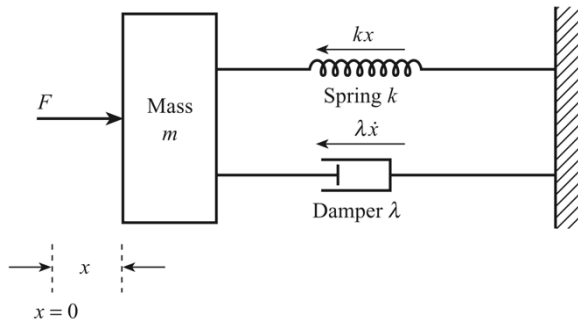
❑ Or $\omega_n = \sqrt{\frac{1}{LC}}$, damping ratio $\zeta = \frac{R}{2L\omega_n}$



$$V = iR + \frac{q}{C} + L \frac{di}{dt}$$

where $i = \frac{dq}{dt}$ (q = charge on the capacitance)

$$\text{thus } L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = V$$



Second order system

□ Nondimensionalized form: $\ddot{x} + 2\omega_n\zeta\dot{x} + \omega_n^2x = \frac{F}{m}$

□ Characteristic equation: $r^2 + 2\omega_n\zeta r + \omega_n^2 = 0 \Rightarrow r = \frac{-2\omega_n\zeta \pm \sqrt{4\omega_n^2\zeta^2 - 4\omega_n^2}}{2} = -\omega_n\zeta \pm \omega_n\sqrt{\zeta^2 - 1}$

□ Solution $\left\{ \begin{array}{l} x = Ae^{-\omega_n\zeta t} \sin(\sqrt{1 - \zeta^2}\omega_n t + \phi), \text{ when } 0 < \zeta < 1, \text{ under damped} \\ x = (A + Bt)e^{-\omega_n\zeta t}, \text{ when } \zeta = 1, \text{ critical damped} \\ x = Ae^{(-\omega_n\zeta + \omega_n\sqrt{\zeta^2 - 1})t} + Be^{(-\omega_n\zeta - \omega_n\sqrt{\zeta^2 - 1})t}, \text{ when } \zeta > 1, \text{ over damped} \end{array} \right.$

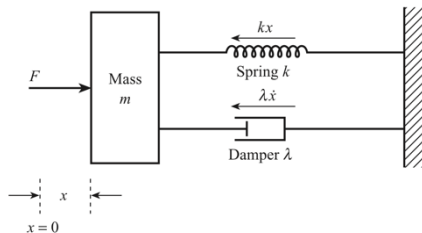
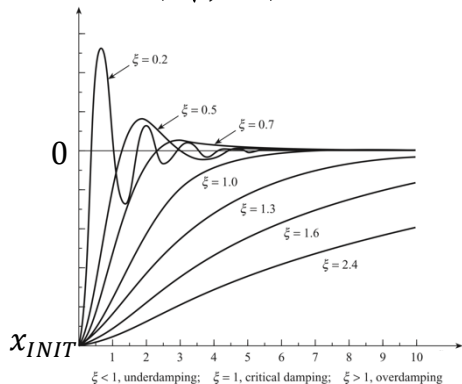


Illustration of 2nd order system

With step response: initial position $x(0) = x_{INIT}$, $\dot{x}(0) = 0$

Solution

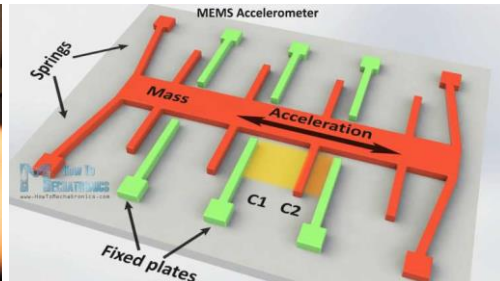
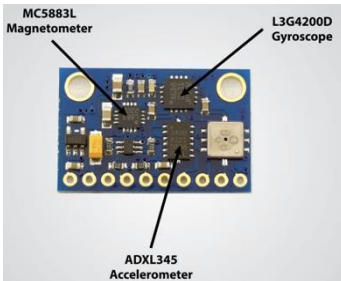
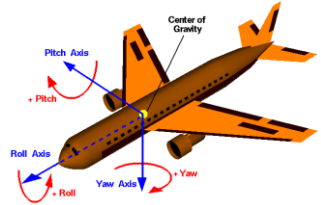
$$\left\{ \begin{array}{l} x = x_{INIT} e^{-\omega_n \zeta t} \sin(\sqrt{1 - \zeta^2} \omega_n t + \phi), \text{ when } 0 < \zeta < 1, \text{ under damped} \\ x = (x_{INIT} + \omega_n \zeta x_{INIT} t) e^{-\omega_n \zeta t}, \text{ when } \zeta = 1, \text{ critical damped} \\ x = \left(\frac{\zeta + \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} \right) x_{INIT} e^{(-\omega_n \zeta + \omega_n \sqrt{\zeta^2 - 1})t} + \left(\frac{\sqrt{\zeta^2 - 1} - \zeta}{2\sqrt{\zeta^2 - 1}} \right) x_{INIT} e^{(-\omega_n \zeta - \omega_n \sqrt{\zeta^2 - 1})t}, \zeta > 1, \text{ over damped} \end{array} \right.$$



Time [s]

The example of IMU

- ❑ An object have 6 degrees of freedom: $x, y, z, \theta_{roll}, \theta_{pitch}, \theta_{yaw}$.
- ❑ An IMU (Inertial Measurement Unit) consists of Magnetometer ($\theta_{roll}, \theta_{pitch}, \theta_{yaw}$), Gyroscope ($\dot{\theta}_{roll}, \dot{\theta}_{pitch}, \dot{\theta}_{yaw}$), and Accelerometer ($\ddot{x}, \ddot{y}, \ddot{z}$)
- ❑ Basically, it is a spring-mass-damper system
- ❑ Enough damping to kill oscillation, but not too much $1 > \zeta > 0.1$
- ❑ Large enough $\omega_n = \sqrt{\frac{k}{m}}$ to make sure respond fast enough.





Exercise

A force sensor has a mass of 0.5 kg, stiffness of $2 \times 10^2 \text{ N m}^{-1}$ and a damping constant of 6.0 N s m^{-1} .

- (a) Calculate the steady-state sensitivity, natural frequency and damping ratio for the sensor.
- (b) Calculate the displacement of the sensor for a steady input force of 2 N.
- (c) If the input force is suddenly increased from 2 to 3 N, derive an expression for the resulting displacement of the sensor.

Laplace transform

□ Laplace transfer: $T_F(s) = \int_0^{\infty} T_F(t) e^{-st} dt$

□ Step: $F(s) = \int_0^{\infty} 1 e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$

□ $\frac{MC}{UA} \dot{T} + T = T_F \Rightarrow$ Transferfunction $G(s) = \frac{T}{T_F} = \frac{1}{\frac{MC}{UA}s + 1}$

□ Response $T(s) = \frac{1}{\frac{MC}{UA}s + 1} \frac{1}{s} = \frac{1}{s} - \frac{\frac{MC}{UA}}{\frac{MC}{UA}s + 1}$

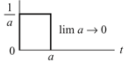
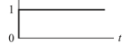
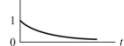
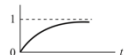
□ Inverse Laplace transfer: $T(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} T(s) e^{st} ds$

□ $T(t) = A - B e^{-\frac{UA}{MC}t}$

□ Apply initial and final conditions, obtain

□ $T(t) = T_F - (T_{INIT} - T_F) e^{-\frac{UA}{MC}t}$

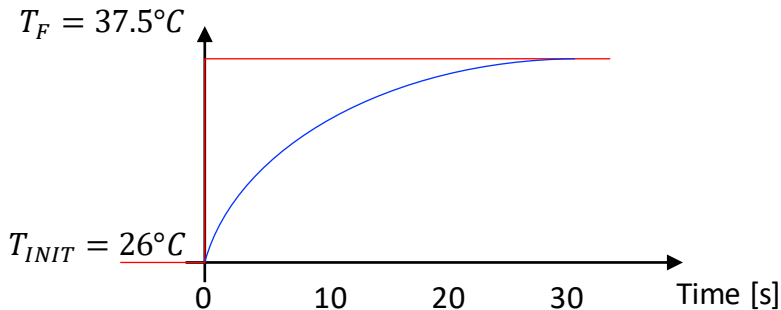
$$\mathcal{L}[f(t)] = f(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Function	Symbol	Graph	Transform
1st derivative	$\frac{d}{dt}f(t)$		$sf(s) - f(0-)$
2nd derivative	$\frac{d^2}{dt^2}f(t)$		$s^2f(s) - sf(0-) - \dot{f}(0-)$
Unit impulse	$\delta(t)$		1
Unit step	$\mu(t)$		$\frac{1}{s}$
Exponential decay	$\exp(-\alpha t)$		$\frac{1}{s + \alpha}$
Exponential growth	$1 - \exp(-\alpha t)$		$\frac{\alpha}{s(s + \alpha)}$

Function, $f(t)$	Laplace transform, $F(s)$	Function, $f(t)$	Laplace transform, $F(s)$
1	$\frac{1}{s}$	$e^{-at} \cos bt$	$\frac{s+a}{(s+a)^2 + b^2}$
t	$\frac{1}{s^2}$	$\sinh bt$	$\frac{b}{s^2 - b^2}$
t^2	$\frac{2}{s^3}$	$\cosh bt$	$\frac{s}{s^2 - b^2}$
t^n	$\frac{n!}{s^{n+1}}$	$e^{-at} \sinh bt$	$\frac{b}{(s+a)^2 - b^2}$
e^{at}	$\frac{1}{s-a}$	$e^{-at} \cosh bt$	$\frac{s+a}{(s+a)^2 - b^2}$
e^{-at}	$\frac{1}{s+a}$	$t \sin bt$	$\frac{2bs}{(s^2 + b^2)^2}$

Laplace transform

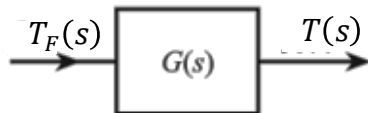
- ❑ $T(t) = 37.5 - 11.5e^{-\frac{UA}{Mc}t}$
- ❑ The same result
- ❑ Another method



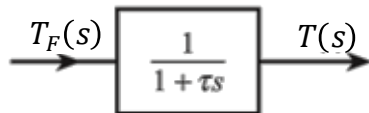
Transfer function

$$\square \frac{MC}{UA} \dot{T} + T = T_F$$

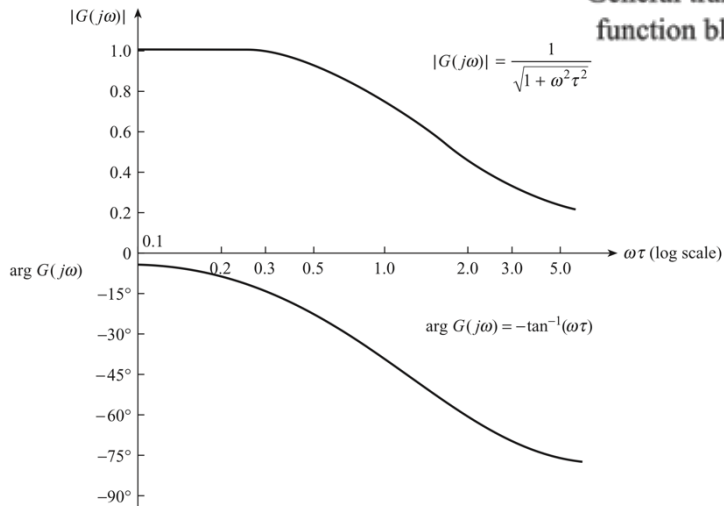
$$\square \Rightarrow G(s) = \frac{T}{T_F} = \frac{1}{\frac{MC}{UA}s + 1} = \frac{1}{1 + \tau s}$$



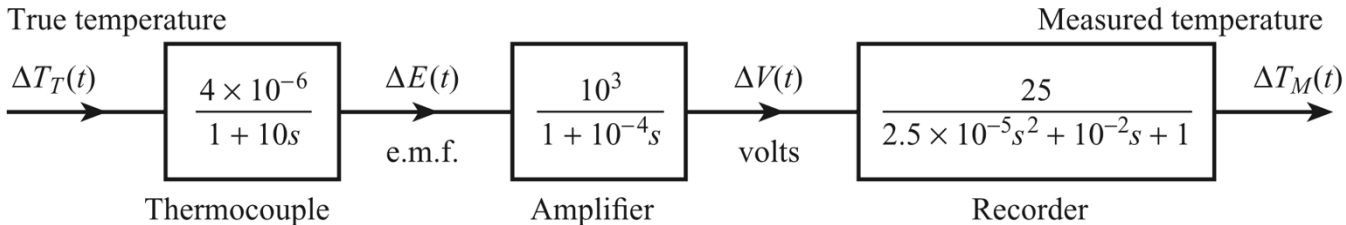
General transfer
function block



First-order
function block



Whole sensor system example



$$\begin{aligned}\Delta \bar{T}_M(s) &= 20 \frac{1}{s} \frac{1}{(1 + 10s)} \frac{1}{(1 + 10^{-4}s)} \frac{1}{(1 + 1/200s)^2} \\ &= 20 \left\{ \frac{1}{s} - \frac{A}{(s + 0.1)} - \frac{B}{(s + 10^4)} - \frac{Cs + D}{(s + 200)^2} \right\}\end{aligned}$$

Using Table 4.1 and eqn [4.30],

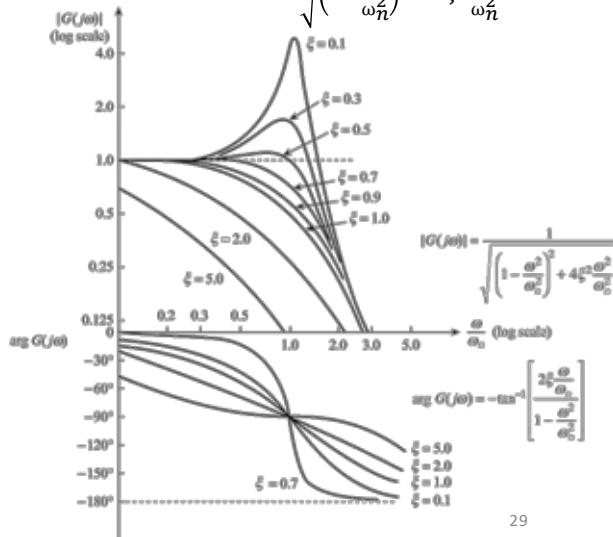
$$\Delta T_M(t) = 20 \{ u(t) - A e^{-0.1t} - B e^{-10^4 t} - E e^{-200t}(1 + 200t) \}$$

Second order system amplitude/phase vs. freq.

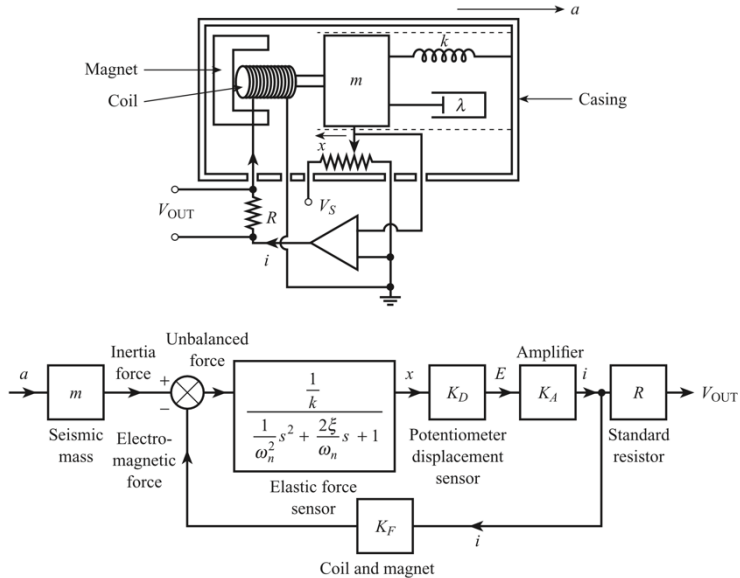
$$\square \ddot{x} + 2\omega_n \zeta \dot{x} + \omega_n^2 x = \frac{F}{m} \Rightarrow \text{Transferfunction } G(s) = \frac{x}{F} = \frac{\frac{1}{m}}{s^2 + 2\omega_n \zeta s + \omega_n^2} = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}}$$

$$\text{Amplitude ratio} = |G(j\omega)| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2 \frac{\omega^2}{\omega_n^2}}}$$

$$\text{Phase difference} = \arg G(j\omega) = -\tan^{-1} \left[\frac{2\zeta \omega / \omega_n}{1 - \omega^2 / \omega_n^2} \right]$$



Tutorial problems



Exercise

4.5 An elastic force sensor has an effective seismic mass of 0.1 kg, a spring stiffness of 10 N m^{-1} and a damping constant of 14 N s m^{-1} .

- (a) Calculate the following quantities:
- (i) sensor natural frequency
 - (ii) sensor damping ratio
 - (iii) transfer function relating displacement and force.
- (b) The above sensor is incorporated into a closed-loop, force balance accelerometer. The following components are also present:

Potentiometer displacement sensor: sensitivity 1.0 V m^{-1}

Amplifier: voltage input, current output, sensitivity 40 A V^{-1}

Coil and magnet: current input, force output, sensitivity 25 N A^{-1}

Resistor: $250 \text{ } \Omega$.

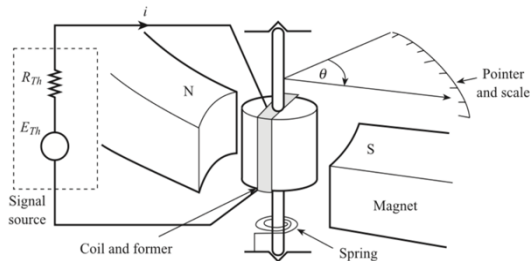
- (i) Draw a block diagram of the accelerometer.
- (ii) Calculate the overall accelerometer transfer function.
- (iii) Explain why the dynamic performance of the accelerometer is superior to that of the elastic sensor.

Lecture 3

- ❑ Preview study : *Principles of Measurement Systems*, Chapter 4: Dynamic Characteristics of Measurement Systems

A moving square-shaped coil indicator is connected to a Thévenin signal source of resistance $12\ \Omega$. Use the data given to answer the following questions.

- What is the sensitivity of the coil indicator?
- What is the natural frequency and damping ratio for the system?
- What is the working frequency range of the system?



Number of turns on coil = 100

Coil resistance = $75\ \Omega$

Coil area = $10^{-4}\ \text{m}^2$

Coil moment of inertia = $2.5 \times 10^{-5}\ \text{kg m}^2$

Magnetic flux density = $150\ \text{Wb m}^{-2}$

Spring stiffness = $10^{-3}\ \text{Nm rad}^{-1}$
